

Introduction to differential games and Hamilton-Jacobi equations

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Abstract

The course is an introduction to the theory of zero-sum differential games (strategies, value function). We will show that the value function of the game satisfies a partial differential equation, called the Hamilton-Jacobi equation, in a weak sense (viscosity solution). We will then discuss the various properties of these viscosity solutions: stability, comparison. We will conclude with the analysis of the long time average of the solution: notion of cell problem and ergodic constant when the state space is compact, or link with first passage percolation theory in stationary ergodic regimes.